

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION TO PROBLEM-SOLUTION FIT

The Problem-Solution Fit phase evaluates how well the proposed product (ComicCraft) addresses the specific pain points of the target audience. Achieving this fit means the product is not just a technological demonstration, but a highly useful tool that resolves a real-world bottleneck. In the context of digital storytelling, the bottleneck is the sheer difficulty of creating visual art without formal training.

3. THE CORE PROBLEM REVISITED

Currently, individuals who possess strong creative writing skills but lack drawing abilities face a massive "Visualization Barrier." To create a comic book, a writer typically needs to hire an illustrator, which is expensive, or learn complex graphic design software, which is incredibly time-consuming. Existing AI image generators (like Midjourney or DALL-E) only generate standalone images. They do not automatically understand a story's narrative flow, nor do they sequence multiple images into a cohesive, structured comic book layout.

4. THE TARGET USER

The target users are amateur storytellers, creative writing students, educators, and hobbyists. They want to instantly see their ideas brought to life in a visual format to share with friends, students, or on social media. They do not want to deal with prompt engineering, image stitching, or manual PDF formatting.

5. THE PROPOSED SOLUTION (COMICCRAFT AI) ComicCraft is an end-to-end, AI-powered web application that automates the entire comic creation pipeline. A user inputs a simple story prompt, character name, setting, and tone. The system uses Google Gemini AI to structure the story into panels and write the dialogue, and

uses Hugging Face Stable Diffusion to generate matching illustrations. Finally, it compiles everything into a downloadable PDF.

6. MAPPING THE SOLUTION TO THE PROBLEM

- **Problem:** Hiring illustrators is too expensive.
 - **Fit:** ComicCraft utilizes automated Generative AI APIs, making the comic creation process practically free and instant.
- **Problem:** AI image tools only generate single, disconnected images.
 - **Fit:** ComicCraft uses Gemini Flash to break the prompt into a 5-panel structured JSON outline, ensuring a continuous narrative flow before generating any images.
- **Problem:** Users struggle to format images and text into a comic page.
 - **Fit:** The application's backend uses a Layout Builder (`layout_builder.py`) and an Exporter (`exporters.py`) utilizing FPDF to dynamically arrange images, scene descriptions, and dialogue into a professional multi-page PDF document.

7. VALUE PROPOSITION AND CONCLUSION ComicCraft delivers immense value by democratizing visual storytelling. It proves a perfect problem-solution fit by completely eliminating the technical, financial, and artistic barriers traditionally associated with comic book creation, providing a seamless "Prompt-to-PDF" pipeline.